

PostgreSQL High Availability Solution

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1 System Overview

1.1 Architecture

- Primary node (read/write)
- 2+ replica nodes (hot standby)
- Automated failover with Bash/Python
- Health monitoring system

1.2 Components

Component	Version
PostgreSQL	14+
Streaming Replication	Native
Failover Manager	Bash Scripts
Monitoring	Custom Python

2 Installation Guide

2.1 Requirements

- 3+ Ubuntu/Debian 20.04/22.04 servers
- SSH key-based access between nodes
- Minimum 4GB RAM, 2 CPU cores per node

2.2 Setup Commands

```
1 sudo NODE_TYPE=primary bash setup_postgres_ha.sh
```

Listing 1: Primary Node Setup

```
1 sudo NODE_TYPE=replica PRIMARY_IP=<primary_ip> bash  
  setup_postgres_ha.sh
```

Listing 2: Replica Node Setup

3 Failover Procedures

3.1 Automated Failover

1. Health checks fail (30s timeout)
2. Trigger file created
3. Replica promotion
4. Cluster reconfiguration

3.2 Manual Failover

```
1 sudo -u postgres pg_ctl promote -D /var/lib/postgresql/14/main
```

Listing 3: Manual Promotion

4 Recovery Processes

4.1 Primary Recovery

```
1 sudo -u postgres pg_rewind -D /var/lib/postgresql/14/main \  
2 --source-server="host=new_primary"
```

Listing 4: Failed Primary Recovery

5 Monitoring System

5.1 Metrics Collected

- Node availability (up/down)
- Replication lag (seconds)
- Disk space usage (%)
- Connection latency (ms)

5.2 Alert Thresholds

Metric	Warning	Critical
Replication Lag	10s	30s
Disk Space	80%	90%

6 Limitations

- Maximum 5 nodes supported
- Failover time 30-60 seconds
- No automatic backup management

7 Future Improvements

7.1 Short-term

- Prometheus metrics endpoint
- Slack notifications

7.2 Long-term

- Kubernetes operator
- Web management UI

8 Maintenance

8.1 Routine Checks

- Daily replication verification
- Monthly failover testing

9 Troubleshooting

9.1 Common Issues

```
1 SELECT pg_wal_replay_resume();
```

Listing 5: High Replication Lag

References

- [1] PostgreSQL Documentation, <https://www.postgresql.org/docs/>